

ECE250: Signals & Systems Quiz-6

Date: 15th October, 2025

Total Marks: 10

Duration: 15 Minutes

Note:

1. Please provide proper mathematical justifications with your answers. No marks will be awarded without a valid justification.
 2. Institute Plagiarism policy are strictly applicable.
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[CO1 CO2 CO3] [Q1]: Consider a discrete-time LTI system whose frequency response is

$$H(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \frac{\pi}{2} \\ 0, & \frac{\pi}{2} < |\omega| < \pi \end{cases}$$

Given the input to this system is

$$x[n] = \sum_{k=-\infty}^{+\infty} \delta[n - 4k],$$

Determine the Fourier series coefficients of the output $y[n]$ and write its expression.

[5 Marks]

[CO1 CO2 CO4] [Q2]: Calculate the Fourier transform of the following signal:

[3 Marks]

$$e^{-2|t-1|}$$

Sketch and label the magnitude.

[2 Marks]